In addition to the online homework, please hand in the following problems:

Problem 1: Compute the outward flux of the following vector fields through the given solids:

a. $\mathbf{F} = [x^2, y^2, z^2]$ where T is the solid cube in the first octant between the planes $x = 1$, $y = 1$, and $z = 1$.

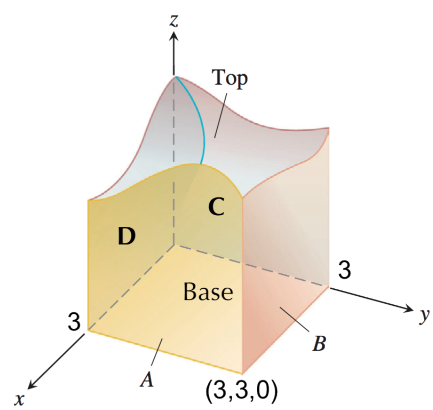
b. $\mathbf{F} = x^3\,\mathbf{i} + y^3\,\mathbf{j} + z^3\,\mathbf{k}$ where T is the solid sphere: $x^2 + y^2 + z^2 \leq 9$

c. $\mathbf{F} = [6x^2 + 2xy,\, 2y + x^2z,\, 4x^2y^3]$ where T is the solid cylinder $x^2 + y^2 = 4$ between the planes $z = 0$ and $z = 3$.

Problem 2: Let $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and suppose that the surface S and region D satisfy the hypotheses of the Divergence Theorem. Show that the volume of D is given by the formula

$$\text{Volume of D} = \frac{1}{3}\iint_S \mathbf{F}\cdot\mathbf{n}\,dA$$

Problem 3: The the closed cube-like surface, S, shown below has a top that is an unknown smooth surface. Let $\mathbf{F} = (5x+2)\mathbf{i} - 7y\mathbf{j} + 7z\mathbf{k}$. Suppose we knew the outward flux through sides C and B were both -4,the volume of the solid is 10 and the area of A is 4. Find the outward flux through the top?



Problem 4: Show that the outward flux of a constant vector field $\mathbf{F}$ across any closed surface is zero.